Statistics for Machine Learning

1. Why statistics matters

Statistics is the toolkit for making sense of data and making decisions when you cannot check every single case yourself. A company cannot ask every customer on earth what they think, a hospital cannot test a new medicine on every human alive, and a government cannot personally interview every citizen. Statistics lets you look at a small chunk of information and still say something trustworthy about the whole picture.

It shows up everywhere: businesses use it to set prices and predict demand, hospitals use it to test if a new medicine actually works better than the old one, governments use it for surveys and election polls, and climate scientists use it to predict future weather from past patterns.

For machine learning specifically, statistics is not just a side skill, it is the actual foundation. Linear Regression, Logistic Regression, SVM and Naive Bayes are all, underneath the code, just statistics formulas wearing a machine learning costume.

2. Descriptive vs Inferential Statistics

Descriptive Statistics

This just describes the data you already have in front of you. No guessing, no prediction, just summarizing. Mean, median, standard deviation, a bar chart, a box plot, all of that is descriptive. It answers: what does my data look like right now.

Inferential Statistics

This uses a small sample of data to make an educated guess about a much bigger group you did not fully measure. It answers: based on this small piece I collected, what is probably true for everyone. This is the harder, more powerful job, and it is where hypothesis testing and confidence intervals live.

- **Example:** calculating the average height of 50 students you measured is descriptive. Using those 50 students to guess the average height of the entire country is inferential.

3. Population, Sample, Parameter and Statistic

These four words are used constantly and people mix them up, so lock these in early.

- **Population:** the entire group you actually care about. Example, every student at a university.
- **Sample:** a smaller slice of the population that you actually collect data from, because measuring the whole population is usually too slow, too expensive, or flat out impossible.
- **Parameter:** a number that describes the population. It is almost always unknown, because you never measured the whole population, you are just trying to estimate it.
- **Statistic:** a number that describes your sample. This one you actually know for sure, because you calculated it yourself from the data you collected.

The entire point of inferential statistics is to use a known statistic from your sample to make a smart guess about the unknown parameter of the population.

What makes a sample trustworthy

A bad sample gives you a confidently wrong answer, which is worse than no answer at all. Three things need to be true:

- Sufficient size: too small a sample and random flukes dominate the result.
- Randomness: everyone or everything in the population needs a fair chance of being picked, otherwise you introduce bias.
- **Representativeness:** the sample needs to contain the same variety as the population. If you want the average salary in India, you cannot only survey people from one city or one profession, you need a mix of states, genders, ages and income classes.

4. Types of Data

Before you calculate anything or plot any graph, you need to know what kind of column you are dealing with. Using the wrong type of statistic on the wrong type of data gives you nonsense results, like calculating an average of city names.

Categorical Data

Data that represents categories or labels, not actual measurable quantities.

- Nominal: categories with no natural order. Example, gender, blood group, city name.
- Ordinal: categories that do have a natural order or ranking. Example, movie rating (1 to 5 stars), education level (school, bachelors, masters).

Numerical Data

Data that represents actual measurable quantities, where math like averaging genuinely makes sense.

- Discrete: countable, whole number values with gaps between them. Example, number of children, number of cars sold.
- Continuous: can take any value including decimals, with no real gaps. Example, height, weight, temperature, salary.

This single decision, categorical or numerical, is what decides everything downstream: which average makes sense, which graph to plot, and which statistical test is valid later on.

5. Measures of Central Tendency

A measure of central tendency is one single number that tries to represent the whole data set. Instead of looking at a thousand numbers, you look at one number that captures where most of the data sits.

Mean (Average)

Formula

Mean = (sum of all values) / (number of values)

Best used when data is roughly balanced with no extreme values. Its big weakness is that it gets pulled hard by outliers.

- **Example:** salaries of 9 people at 40k each and 1 CEO at 5 million. The mean salary looks huge and misleading, it does not represent a typical employee at all.

Median (Middle Value)

Sort all the values and pick the middle one. If there are two middle values, average those two. Median completely ignores how extreme the outliers are, it only cares about position, so it survives the CEO salary problem above untouched.

Mode (Most Frequent Value)

The value that shows up most often. This is the only central tendency measure that works for categorical data, since you cannot calculate a mean or median of colors or city names.

- **Example:** the most common shoe size sold in a store is the mode, useful directly for restocking decisions.

Which one should you actually use

If your numerical data has no crazy outliers, mean is fine and uses all the information. If your data has extreme outliers (salary, house prices), median is safer and more honest. For pure categorical data, mode is your only real option.

6. Measures of Dispersion

Central tendency alone can lie to you. Two classrooms can both have an average mark of 70, but one class might have everyone scoring between 65 and 75, while the other has half the class at 40 and half at 100. Same average, completely different reality. Dispersion measures capture that spread.

Range

Formula

Range = Maximum value - Minimum value

Quick and easy, but a terrible measure on its own because a single crazy outlier can blow it up completely, giving a false impression of how spread out most of the data actually is.

Variance

The average of the squared distance of every point from the mean. Squaring does two jobs at once: it stops positive and negative distances from cancelling each other out, and it punishes bigger deviations more heavily than small ones.

Population Variance

sigma^2 = sum of (Xi - mu)^2 / N

Sample Variance uses (n-1) in the denominator instead of N, and uses the sample mean x-bar instead of mu. This n-1 adjustment corrects for the fact that a sample tends to slightly underestimate the true population spread.

Weakness: squaring means one huge outlier can massively inflate the variance, same problem as range but in a sneakier form.

Mean Absolute Deviation (MAD)

Formula

MAD = sum of |Xi - mean| / N

Uses absolute value instead of squaring.

Less sensitive to ouatliers than variance since it does not square anything. Its downside is that it is mathematically awkward to use for inferential statistics, you cannot cleanly extend a sample's MAD to make claims about the population's MAD, which is why variance remains the more commonly used tool despite being more outlier sensitive.

Standard Deviation

Formula

Standard Deviation = square root of Variance

Variance's units are squared (e.g. rupees squared, marks squared), which does not mean anything intuitive to a human. Taking the square root brings the unit back to something you can actually picture, like rupees or marks. This is why standard deviation, not variance, is the number people actually quote in real life.

Coefficient of Variation (CV)

Formula

CV = (Standard Deviation / Mean) x 100

Standard deviation cannot compare spread across two totally different scales. You cannot fairly compare the spread of salary (measured in lakhs) against the spread of experience (measured in years), the units are just too different. CV solves this by turning spread into a percentage of the mean, giving you a scale free number you can compare across any two data sets.

- **Example:** comparing how consistent two employees' monthly sales are, even if one sells phones worth thousands and the other sells cars worth lakhs, CV puts both on the same percentage footing.

7. Quantiles and Percentiles

Mean and standard deviation give you a summary, but sometimes you want to know exactly where one specific person or value stands compared to everyone else. That is the job of percentiles, they tell you position, not just spread.

What a percentile means

The Nth percentile is the value below which N percent of the data falls. If you scored in the 90th percentile on an exam, 90 percent of people scored below you, and only 10 percent scored above you.

Percentile location formula

$$PL = (P / 100) \times (N + 1)$$

PL is the position in the sorted list, P is the desired percentile, N is total number of observations. If PL is not a whole number, you interpolate between the two nearest values.

Percentile of a value formula

$$P = (X + 0.5Y) / N \times 100$$

X = values strictly below the given value, Y = values exactly equal to it, N = total values. This tells you what percentile a specific score represents.

Quartiles, Deciles and Quintiles

These are just percentiles grouped into fewer chunks, all of them can technically be pulled directly out of percentiles.

- Quartiles: split data into 4 equal parts. Q1 = 25th percentile, Q2 = 50th percentile (this is also the median), Q3 = 75th percentile.
- Deciles: split data into 10 equal parts.
- Quintiles: split data into 5 equal parts.

Important rule: data must always be sorted from smallest to largest before you calculate any of these, and the exact percentile value you calculate might not even exist as a real data point, it is just marking a location.

8. Five Number Summary and Box Plots

This is a fast, five number snapshot of an entire numerical column, so you do not need to stare at a thousand raw values to understand its shape.

- Minimum (0th percentile)
- Q1, First Quartile (25th percentile)
- Median, Q2 (50th percentile)
- Q3, Third Quartile (75th percentile)
- Maximum (100th percentile)

This is exactly what pandas.describe() gives you, and it is genuinely one of the fastest ways to understand a new dataset before doing anything fancy.

Interquartile Range (IQR)

Formula

IQR = Q3 - Q1

This represents the middle 50 percent of your data, ignoring the extreme top and bottom quarters.

Box Plot (Box and Whisker Plot)

A picture of the five number summary. The box itself covers Q1 to Q3 (the IQR), the line inside the box is the median, and lines called whiskers stretch out toward the minimum and maximum, though the whiskers are capped using a rule rather than just going to the true min and max.

Whisker limits

Lower Whisker = Q1 - 1.5 x IQR **Upper Whisker = Q3 + 1.5 x IQR**

Anything beyond these two limits is flagged as an outlier and plotted as a separate dot.

Box plots are one of the fastest ways to spot skewness, compare two groups side by side (like comparing salary across departments), and immediately spot outliers, all without a single calculation.

9. Covariance

Once you have two numerical columns instead of one, you naturally want to know if they move together. Covariance answers exactly that: as one variable goes up, does the other tend to go up too, go down, or do nothing predictable at all.

Population Covariance

Cov(X,Y) = sum of (Xi - mu_X)(Yi - mu_Y) / N

Sample version divides by (n-1) instead and uses sample means.

- Positive covariance: both increase together. Example, years of experience and salary.
- Negative covariance: one goes up as the other goes down. Example, number of backlogs and chance of getting placed.
- Zero or near zero: no consistent linear pattern, they move independently of each other.

The problem with covariance

Covariance tells you the direction of the relationship, but its actual number is meaningless on its own, because the size of covariance is completely tangled up with the scale of your variables. If you simply changed salary from rupees to paisa, the covariance number would explode even though the real relationship between the variables did not change one bit. This makes it impossible to say if a relationship is weak or strong just by looking at the covariance number.

One neat fact worth remembering: if you calculate the covariance of a variable with itself, you get back exactly its variance. Variance is really just a special case of covariance.

10. Correlation

Correlation exists purely to fix covariance's biggest flaw. It keeps the direction information but adds strength information, and it does this by removing the effect of scale entirely.

Formula

r = Cov(X, Y) / (SD(X) x SD(Y))

Dividing by both standard deviations cancels out the units, so correlation always lands somewhere between -1 and 1, no matter what units your original data was in.

- +1: perfect positive linear relationship.
- -1: perfect negative linear relationship.
- 0: no linear relationship at all.

The closer the number is to 1 or -1, the stronger the relationship, the closer to 0, the weaker it is. Because it is scale invariant, meaning rescaling your data does not change the correlation value at all, correlation is the number people actually trust and report, while covariance mostly just exists as a stepping stone used to calculate it.

11. Correlation Does Not Imply Causation

This is one of the most important warnings in all of statistics. Just because two things move together does not mean one is causing the other. There is often a hidden third factor pulling the strings behind both of them.

- **Example 1:** cities with more firefighters also tend to have bigger fires. Firefighters obviously do not cause fires, big fires cause more firefighters to be sent.
- **Example 2, the classic one:** ice cream sales and crime rates rise together across the year. Ice cream does not cause crime. The real hidden cause is hot weather, which makes people buy more ice cream and also makes more people go outside, leading to more opportunities for incidents.

To actually prove causation, not just correlation, you generally need a proper controlled experiment or a randomized trial, correlation from observed data alone is never enough proof by itself.

12. Multivariate Analysis

Sometimes two columns are not enough, and you need to study three or more variables at once. Different situations call for different visual tricks to squeeze extra dimensions onto a flat screen.

- 3D scatter plot: puts three numerical columns on X, Y and Z axes directly.
- **Hue:** colors the points of a normal 2D scatter plot using a categorical column. Example, plotting age against fare, colored by gender, so you see two relationships in one picture.
- **Facet grids:** draws separate side by side plots for each category, instead of cramming everything onto one chart. Example, a separate total bill versus tip scatter plot for lunch and for dinner.
- **Pair plots:** automatically plots every possible pair of columns in a dataset against each other in one big grid, useful for a fast first look at a brand new dataset.
- **Bubble charts:** a 2D scatter plot where the size of each dot represents a third numerical variable. Example, GDP versus population, where bubble size shows literacy rate.

13. Random Variables and Probability Distributions, quick recap

A random variable turns the outcome of a random experiment into a number, so that math like averages and probabilities can actually be applied to it. Discrete random variables take countable separate values (a die roll), continuous random variables can take any value in a range (a person's height).

A probability distribution lists out every possible value of a random variable next to how likely each one is. For a small number of outcomes, like rolling one or two dice, you can just write this as a table. The problem is this stops working once the number of outcomes gets huge, like rolling 10 dice, or becomes flat out impossible once the variable is continuous, since a continuous variable has infinite possible values and you cannot write infinite rows in a table.

The fix is to describe the relationship as a mathematical function you can plug any value into, instead of a fixed list.

- **PMF, Probability Mass Function:** used for discrete variables, gives the exact probability of one specific value.
- **PDF, Probability Density Function:** used for continuous variables. Its Y-axis is not actual probability, it is density, because for a continuous variable the probability of landing on one exact single value is essentially zero. What you actually calculate is the area under the curve between two points, and that area is the real probability of falling in that range.
- **CDF, Cumulative Distribution Function:** gives the probability that the variable is less than or equal to a certain value, basically a running total. It can be built from either a PMF or a PDF, and for continuous variables, integrating the PDF gives you the CDF, while differentiating the CDF gives you back the PDF.

Every distribution also has parameters, tuning knobs that control its shape, location and spread. For example the normal distribution is fully controlled by just two numbers, its mean and its standard deviation. Once you recognize your real data matches a famous, well studied distribution, you instantly get to borrow decades of known math about it instead of starting from scratch.

14. Density Estimation and KDE

Density estimation is simply the process of figuring out the PDF shape for a real dataset you were handed, since real data does not arrive with a formula attached to it.

Parametric Density Estimation

You assume your data follows a known, famous distribution shape (like Normal), then you just calculate that distribution's specific parameters (like mean and standard deviation) from your data. Fast and simple, but only as accurate as how well your assumption actually matches reality.

Non-Parametric Density Estimation

You make zero assumptions about the shape and let the data speak for itself. More flexible and safer when you are not sure your data matches any famous shape, but more computationally heavy and needs more data to be reliable.

Kernel Density Estimation (KDE)

The most popular non-parametric technique. The idea is surprisingly simple once you picture it: place a small bump, usually a mini Gaussian bell curve, centered exactly on top of every single data point you have. Then stack all these little bumps on top of each other and add up their heights everywhere. Where many points cluster together, the bumps pile up high, giving you a peak. Where points are sparse, the bumps barely overlap, giving you a low flat area.

The width of each little bump is called the bandwidth. A tiny bandwidth creates a jagged, spiky, overfit looking curve, while a huge bandwidth smooths everything into a flat blob that hides real detail. Picking a good bandwidth is the whole trick to using KDE well.

15. Uniform Distribution

This is the simplest possible distribution: every outcome is exactly equally likely, there is no favorite value at all, the graph is flat.

- **Example:** rolling a fair die, each face from 1 to 6 has exactly a 1/6 chance, no number is preferred over another.

Continuous Uniform Distribution

Formula

f(x) = 1 / (b - a)   for a <= x <= b

Written as X ~ Uniform(a, b). The height of the flat line is exactly 1/(b-a) so that the total area under it still adds up to 1.

Discrete Uniform Distribution

Same flat idea but with countable steps instead of a smooth range, like a die roll. Mean, median and mode are all equal here, making it perfectly symmetric.

Variance

$$Var = ((b - a + 1)^2 - 1) / 12$$

Where this actually gets used in Machine Learning

- Random weight initialization in neural networks, so no starting value is unfairly favored over another.
- Random sampling, when splitting data into train and test sets fairly.
- Generating pseudo random numbers for simulations and shuffling.

16. Bernoulli and Binomial Distribution

Bernoulli Distribution

The simplest possible random experiment: exactly two outcomes, success (1) or failure (0), and you only run it once. This is the exact mathematical model behind any yes or no question.

PMF

$$P(X=x) = p^x (1-p)^{(1-x)}$$

p is the probability of success, x is either 0 or 1.

- **Example:** one coin toss, one email being spam or not spam, one customer clicking an ad or not.

Binomial Distribution

Bernoulli only handles one single trial. Binomial answers a bigger, more useful question: if you repeat that same yes or no experiment n separate times, how many total successes should you expect.

Formula

$$P(X=x) = C(n,x) \times p^x \times (1-p)^{(n-x)}$$

C(n,x) counts how many different ways you can arrange x successes among n trials.

It needs three conditions to actually apply: a fixed number of trials, only two outcomes per trial, and the same success probability every single trial.

- **Example:** out of 100 people shown a video, how many will actually like it, given each person has roughly a 30 percent chance of liking it individually.

If p is exactly 0.5, the distribution shape is symmetric like a normal curve. If p climbs above 0.5, the whole shape shifts toward the right, and if p drops below 0.5, it shifts left. This distribution is the backbone behind logistic regression and binary classification problems in ML.

17. Normal Distribution (the Bell Curve)

The single most important distribution in all of statistics, because an enormous number of real world things naturally pile up around a middle value with fewer extreme cases on either side: heights, exam scores, measurement errors, blood pressure, and so on.

PDF Formula

f(x) = 1/(sigma x sqrt(2 pi)) x e^(-0.5 x ((x-mu)/sigma)^2)

The front fraction just makes sure the total area equals 1, the exponential part creates the actual bell shape.

It is controlled by exactly two knobs, written as X ~ N(mu, sigma^2). The mean (mu) slides the whole curve left or right, and the standard deviation (sigma) controls how wide or narrow the bell is. A bigger sigma means a flatter, more spread out curve, a smaller sigma means a taller, tighter curve.

Standard Normal Variate (Z-score)

Every normal distribution has different mu and sigma, which makes comparing two of them directly awkward. The fix is to convert any point into a Z-score, which tells you how many standard deviations away from the mean that point sits, regardless of the original scale.

Formula

Z = (X - mu) / sigma

Once converted, you land on the standard normal distribution, which always has mean 0 and standard deviation 1. This lets you use one single, universal Z-table to look up probabilities for any normal distribution on earth, instead of needing a separate table for every possible mu and sigma combination.

The Empirical Rule (68-95-99.7 Rule)

- About 68 percent of data falls within 1 standard deviation of the mean.
- About 95 percent falls within 2 standard deviations.
- About 99.7 percent falls within 3 standard deviations.

This rule is a fast, no-calculation way to spot outliers, anything sitting beyond 3 standard deviations from the mean is unusual enough to deserve a second look.

Skewness

Skewness measures how lopsided a distribution is. A perfect normal curve is symmetric and has skewness of exactly zero.

- **Positive skew (right skewed):** a longer tail stretches out to the right, like income data, most people earn moderate amounts but a few earn extremely high amounts pulling the tail right. Here Mode < Median < Mean.
- **Negative skew (left skewed):** a longer tail stretches out to the left. Here Mean < Median < Mode.

As a rough guide, skewness between -0.5 and 0.5 is close enough to symmetric, between 0.5 and 1 (or -1 and -0.5) is moderately skewed, and beyond 1 or -1 is heavily skewed and should not be treated as normal. This matters a lot because many ML models and statistical tests silently assume your data is roughly normal, and skipping this check can quietly wreck your results.

CDF of Normal Distribution

Shows the running total probability up to a point, and it always looks like a stretched S shape, starting near 0, crossing 0.5 exactly at the mean, and climbing toward 1 as x grows large.

Why the Normal Distribution matters so much for ML

- Outlier detection, using the empirical rule as a fast check.
- Many models like Linear and Logistic Regression assume the errors they make are normally distributed for their math to work properly.
- Most classical hypothesis tests (t-tests, ANOVA) assume normality underneath.
- **The Central Limit Theorem:** explains why the normal distribution keeps showing up everywhere, covered fully in section 19.

18. Log-Normal Distribution and Transformations

Not everything in the world is naturally symmetric like the normal curve. Income, house prices, and app data usage all tend to cluster at lower values with a long tail of a few extremely high outliers. Forcing normal-distribution assumptions onto data like this gives broken results.

A log-normal distribution is what you get when the logarithm of your variable, not the variable itself, is normally distributed.

> **Definition**
>
> **If Y = log(X) is normally distributed, then X is log-normally distributed.**

This matters because it explains data that can never go negative (you cannot earn negative income) and is skewed right (most people earn a modest amount, a few earn a huge amount).

Checking and fixing skewed data with transformations

If your data looks log-normal or otherwise skewed, and you need it to behave more normally for a model or test to work properly, you apply a transformation.

- **Log transformation:** the most common fix, directly reverses a log-normal shape back toward normal, and helps stabilize variance for linear models.
- **Reciprocal transformation:** flips values (1/x), useful for certain kinds of heavy skew.
- **Square root transformation:** a gentler fix than log, often used on count data.

You check if the fix worked simply by re-plotting the transformed data, or running a normality test on it, and seeing if it now looks close to a bell curve.

19. Sampling Distribution and Central Limit Theorem

This is arguably the most powerful idea in this entire note. It is the reason you are allowed to trust a small sample to make claims about a giant population.

Sampling Distribution

Imagine you do not just take one sample, you take many separate random samples from the same population, and calculate the mean of each one. If you then plot all of those sample means together, that plot is called the sampling distribution of the mean.

- **Example:** draw 100 different samples of 50 people each from a country's population, calculate the average salary in each of those 100 samples, and plot all 100 averages. That plot is a sampling distribution.

Central Limit Theorem (CLT)

This is the actual magic trick. No matter what shape the original population distribution is, skewed, uniform, weird and lumpy, or anything else, if you take a big enough sample (a common rule of thumb is at least 30), the distribution of sample means will still come out looking approximately normal.

Variance of sample means

Var(X-bar) = sigma^2 / n

The sampling distribution's mean equals the population mean, but its spread shrinks as your sample size n grows, meaning bigger samples give more reliable, tighter estimates.

Why this matters so much: it means you can use all the well understood, well studied math of the normal distribution to make confident statements about a population's mean, even if you have no idea what shape the actual population data follows. This single theorem is the reason confidence intervals and hypothesis tests, covered next, are even mathematically allowed to work.

20. Confidence Intervals

A single sample mean is just one guess, called a point estimate, and a single guess never tells you how confident you should be in it. A confidence interval fixes this by giving you a range of values instead of one number, along with a stated confidence level for that range.

What a confidence interval actually means

Saying you are 95 percent confident that the true average lies between 25 and 32 does not mean there is a 95 percent chance the true value is in that specific range, the true population value is a fixed number, it does not move around randomly. What it really means is: if you repeated this entire sampling process many times and built an interval every time, about 95 percent of those intervals would end up containing the true value. This is a commonly misunderstood point, worth remembering exactly as stated.

Z-Procedure (when population standard deviation is known)

Formula

x-bar +/- Z(alpha/2) x (sigma / sqrt(n))

Z(alpha/2) is 1.96 for a 95 percent confidence level. Requires the population standard deviation to be known and either a normal population or a large enough sample size for CLT to kick in.

T-Procedure (when population standard deviation is unknown)

In real life you almost never actually know the true population standard deviation, so you estimate it using your sample's own standard deviation instead. But using an estimated number instead of a true number adds extra uncertainty, so you cannot just reuse the normal distribution, you need the t-distribution instead, which is deliberately a bit wider to account for that extra guesswork.

Formula

x-bar +/- t(alpha/2, df) x (s / sqrt(n))

df stands for degrees of freedom, usually n-1. As sample size grows, the t-distribution gradually becomes almost identical to the normal distribution.

What changes the width of the interval

- Higher confidence level (say 99 percent instead of 95) makes the interval wider, since you are demanding to be more sure.
- More variability in the data (higher standard deviation) makes the interval wider.
- **Bigger sample size makes the interval narrower,** following an inverse square root relationship, meaning you need to quadruple your sample size just to cut your margin of error in half, a classic case of diminishing returns.

Margin of Error

E is proportional to 1 / sqrt(n)

21. Hypothesis Testing Basics

Hypothesis testing is how you use sample data to make a formal, defendable decision about a claim, instead of just eyeballing numbers and guessing. Think of it exactly like a courtroom trial.

Null and Alternate Hypothesis

- **Null Hypothesis (H0):** the boring default assumption, presumed true until proven otherwise, like presumption of innocence in court. Example, a chips packet weighs exactly 100 grams.
- **Alternate Hypothesis (H1):** the claim you are actually trying to find evidence for, like the prosecution trying to prove guilt. Example, the packet does not actually weigh 100 grams.

The two are mutually exclusive, only one can be true, and your job is to use the sample evidence to decide whether there is enough proof to reject the null hypothesis in favor of the alternate.

Steps in Hypothesis Testing

- Formulate the null and alternate hypotheses clearly.
- Pick a significance level alpha, commonly 0.05 or 0.01.
- Check assumptions, like whether the data is normal and whether the population standard deviation is known, to decide between a Z-test or T-test.
- Calculate the test statistic (a Z-value or T-value) from your sample.
- Define the rejection region, critical cutoff values based on alpha.
- Compare your calculated statistic against that rejection region.
- Interpret the result in plain, real world language.

Significance Level and the Two Types of Errors

Alpha is the probability you are willing to accept of making a false alarm, rejecting a null hypothesis that was actually true.

- **Type I Error (false positive):** you reject H0 when it was actually true. Example, concluding a fair coin is biased when it really was not.
- **Type II Error (false negative):** you fail to reject H0 when it was actually false. Example, missing a real, genuine improvement because your evidence was not strong enough.

These two errors trade off against each other, tightening your criteria to avoid one type of mistake makes you more likely to make the other, there is no free lunch here.

One-Tailed vs Two-Tailed Tests

- **Two-tailed test:** checks for any difference at all, in either direction. Used when your claim uses "not equal to".
- **One-tailed test:** checks for a difference in one specific direction only. Used when your claim uses "greater than" or "less than". It is more powerful at detecting an effect in that specific direction, but it is blind to an effect in the opposite direction.

Limitation of the Rejection Region Approach

This approach only gives you a strict yes or no answer, it crosses the line or it does not, and it hides how strong or weak the actual evidence really was. A result that barely crosses the line gets treated exactly the same as one that smashes through it. This exact gap is what the p-value approach was built to fix.

22. The P-Value Approach

The p-value is the probability of seeing a result as extreme as, or more extreme than, what you actually observed, assuming the null hypothesis is true. In plain words, it answers: if nothing unusual was actually going on, how surprising is this result really.

Coin toss example to build intuition

Flip a coin 100 times. If it is truly fair, getting close to 50 heads is expected and not surprising at all, so the p-value would be large. But getting 80 heads out of 100 would be shockingly unlikely for a fair coin, so the p-value would be tiny. A tiny p-value is strong evidence that your assumption (fair coin) is probably wrong.

Decision rule

- p-value less than 0.01: very strong evidence against H0, reject it.
- p-value between 0.01 and 0.05: moderate evidence, reject H0.
- p-value between 0.05 and 0.1: weak evidence, needs more investigation or domain judgment.
- p-value above 0.1: no real evidence against H0, fail to reject it.

The standard shortcut most people actually use in practice is simpler: if p-value is less than alpha (usually 0.05), reject H0, otherwise fail to reject it. Failing to reject H0 never proves H0 is true, it only means you did not gather enough evidence to knock it down, exactly like a not-guilty verdict does not prove innocence, it just means the evidence was not strong enough to convict.

Why p-value beats the plain rejection region approach

A p-value of 0.04 and a p-value of 0.00002 both lead to the exact same reject decision under a 0.05 cutoff, but they represent wildly different strengths of evidence. The p-value approach preserves that extra nuance, which the plain rejection region approach completely throws away.

Worked Example: one-tailed Z-test

A training program is tested to see if it increased productivity. Before training the average was 50 units per day. A sample of 30 employees after training averaged 53 units, with a known population standard deviation of 8.

Z-Statistic

Z = (x-bar - mu) / (sigma / sqrt(n))

Here this works out to Z = 4.10, a very large value.

Since this is a one-tailed test (checking only for an increase), the p-value is the area to the right of Z = 4.10, which comes out to about 0.00002. Since that is far smaller than alpha 0.05, H0 is rejected, the training program genuinely appears to have increased productivity.

Worked Example: two-tailed Z-test

A snack company claims its packets average 50 grams. A watchdog samples 40 packets and finds an average of 49 grams, with a known population standard deviation of 5.

Z-Statistic

$$Z = (49 - 50) / (5 / \mathrm{sqrt}(40)) \approx -1.26$$

Since this test checks for any difference at all (not just higher or lower), it is two-tailed, so you double the one-sided area. The area to the left of -1.26 is about 0.103, so the p-value is 2 x 0.103 = 0.206. Since 0.206 is bigger than 0.05, you fail to reject H0, there is not enough evidence the packet weight is actually off from the claimed 50 grams.

Quick rule to remember: claims using "greater than" or "less than" need a one-tailed test, claims using "not equal to" need a two-tailed test.

23. T-Tests

Z-tests need you to already know the true population standard deviation, which almost never happens in the real world. T-tests solve this by letting you use your sample's own standard deviation instead, at the cost of using the slightly wider t-distribution to account for that extra uncertainty.

Shared assumptions across all t-tests

- The data (or the differences, for paired tests) should be roughly normally distributed, checked with a Shapiro-Wilk test if the sample is small. A Shapiro-Wilk p-value above 0.05 means normality is a safe assumption.
- Observations must be independent of each other.
- Sampling must be random.

One-Sample T-Test

Compares your one sample's mean against a fixed, hypothesized value.

> **Formula**
>
> **T = (x-bar - mu) / (s / sqrt(n))**
>
> *Degrees of freedom = n - 1.*

- **Example:** testing if a new chocolate bar's average weight really is 50 grams, using just one sample and one target value.

Independent Two-Sample T-Test (Unpaired)

Compares the means of two completely separate, unrelated groups, where a person in one group has nothing to do with anyone in the other group.

> **Formula**
>
> **T = [(x-bar1 - x-bar2) - (mu1 - mu2)] / (sp x sqrt(1/n1 + 1/n2))**
>
> *sp is the pooled standard deviation, degrees of freedom = n1 + n2 - 2.*

- **Example:** comparing average screen time between male and female users, two entirely different groups of people.

This test additionally assumes the two groups have roughly equal variances, checked using Levene's Test. A Levene's test p-value above 0.05 means it is safe to assume equal variances.

Paired Two-Sample T-Test (Dependent)

Used specifically when the two sets of numbers are linked to the exact same subjects, most commonly a before-and-after measurement on the same people.

- **Example:** weighing the same 15 people before and after an 8 week weight loss program. Since it is literally the same person in both measurements, this is not two independent groups, it is paired data.

The trick here is simple: calculate the difference for every single person (before minus after), then run essentially a one-sample t-test on those differences, checking if the average difference is meaningfully different from zero.

- **Worked example:** if that weight loss study produces a p-value of 0.207 against alpha 0.05, since 0.207 is bigger than 0.05, you fail to reject H0, there is not enough evidence the program caused real weight loss.

When you have more than two groups: ANOVA

A t-test only ever compares two groups at a time. The moment your categorical variable has three or more categories, like comparing test scores across three different teaching methods, a t-test is the wrong tool. That situation calls for ANOVA, Analysis of Variance, which extends this same hypothesis testing logic to handle many groups at once.

End of notes. If you are short on time before revision, prioritize section 10 (correlation), section 17 (normal distribution), section 19 (CLT) and section 22 (p-value), these four ideas quietly hold up almost everything else in this document.